

REASONING & DECISION MAKING

Types, Patterns & Approaches to Reasoning · Decision Making · Biases in Decision Making

A structured overview of how individuals draw conclusions, evaluate alternatives, and the systematic biases that shape everyday judgment.

- ✓ Meaning & Characteristics of Reasoning
- ✓ Types of Reasoning
- ✓ Patterns of Reasoning
- ✓ Approaches to Reasoning
- ✓ Decision Making & Its Stages
- ✓ Biases in Decision Making

Meaning of Reasoning

Reasoning is a higher-order cognitive process through which individuals use available information, knowledge, and evidence to arrive at conclusions, solve problems, make judgments, and predict outcomes. It involves mental manipulation of concepts and propositions to derive new knowledge from existing information.

Reasoning enables human beings to move beyond immediate experiences and helps in understanding relationships, drawing inferences, and planning future actions.

Characteristics of Reasoning

- It is goal-directed.
- It involves logical thinking.
- It depends upon prior knowledge and experience.
- It helps in problem-solving and decision-making.
- It may be conscious and deliberate or automatic and intuitive.

1. Types of Reasoning

Psychologists generally classify reasoning into several types.

A Deductive Reasoning

Deductive reasoning involves moving from general principles or premises to a specific conclusion. If the premises are true and the reasoning process is valid, the conclusion must necessarily be true.

Structure: General Rule → Specific Case → Conclusion

Example: Premise 1: All mammals breathe air. Premise 2: A whale is a mammal. Conclusion: Therefore, a whale breathes air.

Characteristics: Produces logically certain conclusions; used extensively in mathematics and formal logic; accuracy depends upon the validity of premises.

Advantages

- Provides certainty.
- Useful in scientific hypothesis testing.

Limitations

- Incorrect premises lead to false conclusions.
- Does not generate new knowledge beyond existing assumptions.

B Inductive Reasoning

Inductive reasoning proceeds from specific observations to broader generalizations. The conclusions obtained are probable rather than certain.

Structure: Specific Observations → Pattern Recognition → General Conclusion

Example: Observation 1: Iron expands when heated. Observation 2: Copper expands when heated. Observation 3: Aluminium expands when heated. Conclusion: Metals expand when heated.

Characteristics: Conclusions are tentative; widely used in scientific research; helps formulate theories.

Advantages

- Generates new knowledge.
- Useful for discovering patterns.

Limitations

- Conclusions may be inaccurate if observations are insufficient.
- Subject to biases.

C Analogical Reasoning

Analogical reasoning involves drawing conclusions based on similarities between two situations.

Example: The structure of the atom was understood by comparing it with the solar system.

Features: Depends on perceived similarities; frequently used in teaching and problem-solving; facilitates transfer of learning.

Limitations: Superficial similarities may produce incorrect conclusions.

D Abductive Reasoning

Abductive reasoning seeks the most plausible explanation for an observation.

Example: Observation: The grass is wet. Possible explanations: It rained · Sprinklers were used. Most plausible explanation: It rained.

Characteristics: Common in medical diagnosis; used in detective investigations; conclusions remain tentative.

2. Patterns of Reasoning

Reasoning often follows identifiable patterns.

2.1 Linear Reasoning

Thoughts progress in a sequential manner from one step to another.

Example: Poor sleep → Fatigue → Reduced concentration → Lower academic performance.

2.2 Conditional Reasoning

Uses "if-then" relationships.

Example: If a person studies regularly, then examination performance improves.

Forms:

Modus Ponens

If P, then Q. P is true. Therefore, Q is true.

Example: If it rains, roads become wet. It is raining. Therefore, roads are wet.

Modus Tollens

If P, then Q. Q is not true. Therefore, P is not true.

Example: If the light is on, electricity is available. Electricity is absent. Therefore, the light cannot be on.

2.3 Categorical Reasoning

Based on relationships among categories.

Example: All psychologists are researchers. Some teachers are psychologists. Therefore, some teachers are researchers.

2.4 Probabilistic Reasoning

Decisions are made considering probabilities and uncertainties.

Example: Weather forecasting relies heavily on probabilistic reasoning.

3. Approaches to Reasoning

Psychologists have proposed different perspectives to understand reasoning.

A Logical Approach

Reasoning is viewed as adherence to formal rules of logic.

Assumptions: Individuals think rationally; conclusions follow established logical principles.

Contribution: Explains ideal human reasoning.

Limitation: People frequently violate logical principles in everyday situations.

B Information Processing Approach

This approach considers reasoning as a process involving: encoding information, storage in memory, retrieval, evaluation, and response selection.

Reasoning is influenced by attention, working memory capacity, and cognitive resources.

Contribution: Explains how mental processes operate during reasoning.

C Heuristic Approach

People often use mental shortcuts or heuristics to simplify reasoning. Examples include the availability heuristic, the representativeness heuristic, and the anchoring heuristic.

Advantages

- Saves time and cognitive effort.
- Useful under conditions of uncertainty.

Limitations

- Can lead to systematic errors.

D Dual-Process Approach

This perspective proposes two systems of thinking.

System	Characteristics	Example
System 1	Fast, automatic, intuitive, emotional	Immediately recognizing a familiar face.
System 2	Slow, deliberate, analytical, effortful	Solving a complex mathematical problem.

Most human reasoning involves interaction between these two systems.

Decision Making

Concept of Decision Making

Decision making refers to the cognitive process of selecting one course of action from among several available alternatives to achieve desired goals. It is an essential component of everyday functioning and professional activities.

Characteristics

- Goal-oriented process.
- Requires evaluation of alternatives.
- Involves prediction of outcomes.
- Influenced by emotions, experiences, and social factors.
- May involve risk and uncertainty.

Stages of Decision Making

- 1 Identifying the Problem** — Recognizing the need for a decision.
Example: Choosing a postgraduate specialization.
- 2 Gathering Information** — Collecting relevant data and alternatives.
- 3 Evaluating Alternatives** — Comparing advantages and disadvantages.
- 4 Selecting the Best Option** — Choosing the most suitable course of action.
- 5 Implementing the Decision** — Putting the selected option into practice.
- 6 Reviewing Outcomes** — Assessing effectiveness and making adjustments if needed.

Biases in Decision Making

Decision-making is often affected by systematic cognitive biases.

Bias	Description	Example
1. Confirmation Bias	People seek information that supports existing beliefs while ignoring contradictory evidence.	Reading only articles that support one's political opinions.
2. Availability Bias	Judgments are influenced by information that is easily recalled.	After hearing news about an airplane crash, people may overestimate the danger of flying.
3. Representativeness Bias	Individuals judge probabilities based on similarity to stereotypes.	Assuming a quiet person is more likely to be a librarian than a salesperson.
4. Anchoring Bias	People rely heavily on the first piece of information encountered.	A product initially priced at ₹10,000 may seem inexpensive when discounted to ₹7,000.
5. Overconfidence Bias	People tend to overestimate the accuracy of their judgments.	A student believing they will excel in an examination despite inadequate preparation.
6. Hindsight Bias	The tendency to believe that past events were predictable after they have occurred.	Saying, "I knew this team would win," after the match result is announced.
7. Framing Effect	Choices are influenced by how information is presented.	People may prefer a treatment described as having a 90% survival rate over one described as having a 10% mortality rate, even though both convey the same information.

Conclusion

Reasoning and decision making are central aspects of human cognition. Reasoning enables individuals to derive conclusions, solve problems, and understand relationships, while decision making involves selecting the most appropriate action among alternatives. Although humans strive to make rational decisions, cognitive biases and heuristic shortcuts frequently influence judgments. Understanding these processes helps improve critical thinking, problem-solving skills, and the quality of everyday decisions.